

Combined Quantum and Post-Quantum Security for Earth-Satellite Channels

Anju Rani et al. (IEEE International Conference on Quantum Communications, Networking, and Computing 2025)

July 10, 2025

1 Introduction

2 Main

- Combining QKD and AES-256
- EPS and Entangled Photons
- Photon Path: Steps 1–4

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- Secure communication is critical in the era of emerging quantum computers.
- Classical cryptography may be broken by future quantum attacks.
- Two promising approaches:
 - Quantum Key Distribution (QKD)
 - Post-Quantum Cryptography (PQC)
- This paper demonstrates a real-time prototype combining QKD and PQC.

- Implement an **entanglement-based QKD (BBM92) system** in real time.
- Mimic an **Earth-satellite channel** in a lab environment.
- Combine QKD with **AES-256** to ensure security even if one primitive fails.
- Demonstrate feasibility of combined QKD-PQC for secure key exchange.

Why Both QKD and PQC?

- Both aim to securely establish a shared secret key.
- But they come from very different paradigms:

Property	QKD	PQC
Origin	Physics	Mathematics
Security	Information-theoretic	Computational
Requires Hardware?	Yes (photons, detectors)	No (CPUs only)
Cost	High	Low
Key Rate	Low	High
Scalability	Difficult	Easier

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Overview

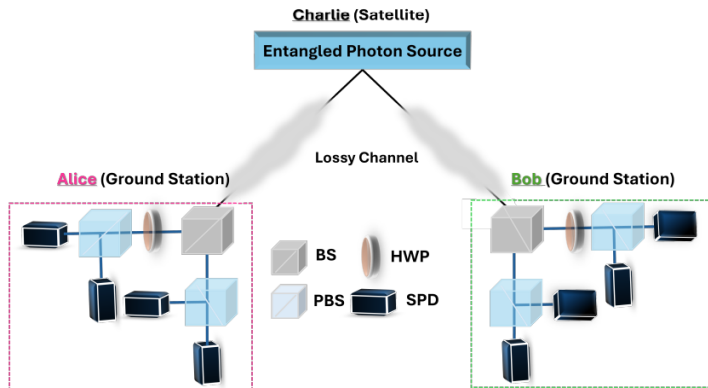


Fig. 1. Our entanglement-based satellite quantum communication system. HWP: Half-wave plate; PBS: Polarizing beam splitter; SPD: Single-photon detector; BS: Beam splitter.

Why Combine QKD and AES-256?

- QKD alone:
 - Generates secret keys using quantum physics.
 - Keys are information-theoretically secure.
 - But: slow key rate, and key length may not suffice for large messages.
- AES-256:
 - Uses pre-shared 256-bit key (symmetric).
 - Fast and efficient, computationally secure even against quantum attacks.
- Combining both provides **defense in depth**.

How Are Keys Used?

- After QKD:
 - If QKD key $\geq$ message length: use QKD key directly.
 - If QKD key $<$ message length: fall back to AES-256.
- AES-256 key:
 - Pre-shared between Alice and Bob.
 - Chosen from part of a secure, pre-agreed key material.
- Both message and **syndrome messages** (from QKD error correction) are encrypted with AES-256 for additional protection.

Why Not Hybrid Keys?

- Some prior works propose **hybrid keys**:
 - Combine QKD and PQC keys (e.g., via XOR).
 - Requires keys to have equal length.
- Authors chose **not** to use hybrid keys:
 - QKD key rate too low to match PQC key length.
 - Hybrid keys raise compositional security concerns.
 - Their approach avoids these burdens.

- Layered design:
 - QKD: information-theoretic security.
 - AES-256: computational security.
- Even if one is broken:
 - If QKD fails $\rightarrow$ AES still protects.
 - If AES fails $\rightarrow$ QKD still protects.
- Only if both are broken simultaneously does security fail.

What is a Syndrome Message?

- During QKD, Alice and Bob obtain raw keys that are similar but not identical due to noise.
- They perform **information reconciliation** to correct errors using error-correcting codes.
- Alice computes the **syndrome** of her bitstring and sends it to Bob.

Definition:

$$s = Hx$$

- x : Alice's raw key.
- H : parity-check matrix of an error-correcting code.
- s : syndrome — compact information about x that helps Bob correct his key.

Example:

$$x = (1\ 0\ 1\ 1\ 1\ 0), \quad H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$
$$s = Hx \mod 2 = (1\ 1)$$

- Bob uses y (his noisy string) and s to correct his bits.

Step 1: What does the EPS generate?

- The Entangled Photon Source (EPS) produces pairs of photons via:
 - A nonlinear crystal (ppKTP) pumped by a laser.
 - Process called **Spontaneous Parametric Down-Conversion (SPDC)**.
- One high-energy photon splits into two lower-energy photons.
- These two photons are created simultaneously and are entangled.

Step 2: What are “Signal” and “Idler”?

- The two photons from EPS are labeled:
 - **Signal photon** — sent to Alice.
 - **Idler photon** — sent to Bob.
- These are just labels: they are symmetric and equally important.
- Both photons carry polarization information and are entangled.

Step 3: Are the photons polarized?

- Yes — the two photons are emitted with orthogonal polarizations:
 - One is horizontally polarized (H).
 - The other is vertically polarized (V).
- The exact outcome isn't fixed — they exist in a superposition until measured.

Step 4: The entangled state

- The state of the two photons is described by:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle_s|V\rangle_i + |V\rangle_s|H\rangle_i)$$

- This is a **Bell state** — a maximally entangled two-qubit state.
- Interpretation:
 - If Alice measures H , Bob will measure V .
 - If Alice measures V , Bob will measure H .
- Neither photon has a definite polarization on its own.

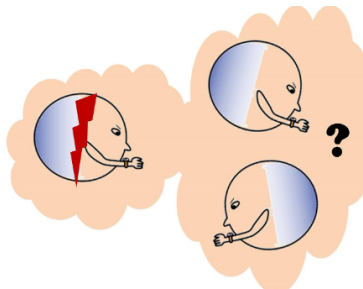
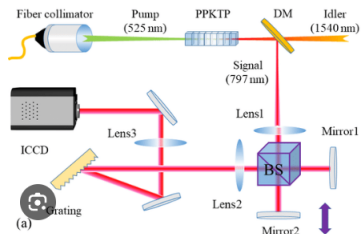
Summary: EPS and Entangled Photons

- The Entangled Photon Source (EPS) generates pairs of photons using:
 - A high-energy laser (pump) at specific wavelength.
 - A nonlinear crystal (ppKTP) via **Spontaneous Parametric Down-Conversion (SPDC)**.
- In SPDC:
 - One pump photon splits into two lower-energy photons.
 - The two photons are created simultaneously and are entangled.
- The two photons are called:
 - **Signal photon** — sent to Alice.
 - **Idler photon** — sent to Bob.
- The two photons are entangled in their **polarization states**:
 - One is horizontally (H) polarized.
 - The other is vertically (V) polarized.
 - But which is which is undetermined until measured.
- The entangled state is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle_s|V\rangle_i + |V\rangle_s|H\rangle_i)$$

- Measurement outcomes are perfectly correlated:

Illustration in Figs



Overview of the System

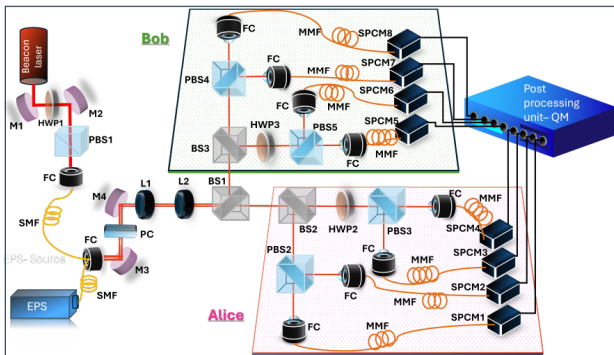


Fig. 2. The experimental setup for the BBM92 protocol. EPS: Entangled photon source; M: Mirror; HWP: Half-wave plate; PBS: Polarizing beam splitter; FC: Fiber coupler; SMF: Single-mode fiber; BS: Beam splitter; PC: Polarization correction optics which is a combination of a quarter-wave plate - half-wave plate - quarter-wave plate; L: Lens; MMF: Multi-mode fiber; SPCM: Single-photon counting module; QM: Quantum machine (post-processing unit).

Step 1: Entangled Photon Generation

- The EPS generates a pair of entangled photons using:
 - A high-energy laser (pump) at specific wavelength.
 - A nonlinear crystal (ppKTP) via **Spontaneous Parametric Down-Conversion (SPDC)**.
- One pump photon splits into two lower-energy photons: the **signal** and the **idler**.
- These two photons are entangled in their polarization:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|H\rangle_s |V\rangle_i + |V\rangle_s |H\rangle_i)$$

Step 2: Fiber Coupling and Free Space

- The signal and idler photons initially co-propagate through a single-mode fiber (SMF).
- At the fiber end, a **fiber coupler (FC)** launches the photons into free space.
- In the lab:
 - Mirrors (M3, M4) redirect the beam to fit the optical table.
 - Lenses (L1, L2) collimate and focus the beam for correct propagation.
- This simulates a satellite-to-ground optical channel in the laboratory.

Step 3: Polarization Drift and Correction

- Traveling through fiber can cause **polarization drift**:
 - Fiber imperfections and stress rotate the polarization states unpredictably.
- To correct this drift:
 - Photons pass through a set of wave plates: **QWP – HWP – QWP**.
 - The angles of these plates are adjusted to restore the intended polarization.

Step 4: Beam Splitter $\rightarrow$ Alice & Bob

- After polarization correction and proper beam shaping:
- The photons reach a **50:50 beam splitter (BS1)**.
- The beam splitter randomly directs:
 - One photon to Alice.
 - The other photon to Bob.
- This mimics a satellite splitting the entangled pair and sending each photon to different ground stations.

Step 5: Measurement at Alice & Bob

- After BS1 splits the entangled photons:
 - One photon goes to Alice.
 - The other photon goes to Bob.
- Goal: Measure the polarization state of the incoming photon.
- Two possible bases:
 - $\{\mathbf{H}, \mathbf{V}\}$: Horizontal / Vertical.
 - $\{\mathbf{D}, \mathbf{A}\}$: Diagonal / Anti-diagonal.

Random Basis Choice

- At both Alice & Bob:
 - A local 50:50 beam splitter randomly selects the measurement basis.
- Outcome:
 - One path $\rightarrow$ measure in $\{H, V\}$.
 - Other path $\rightarrow$ measure in $\{D, A\}$.
- Ensures security by making basis choice unpredictable.

Measuring $\{H, V\}$ Basis

- If path selects $\{H, V\}$:
 - Photon enters a Polarizing Beam Splitter (PBS).
 - PBS splits photon into H and V components.
 - Alice: uses PBS2.
 - Bob: uses PBS4.

Measuring $\{D, A\}$ Basis

- If path selects $\{D, A\}$:
 - Photon passes through a Half-Wave Plate (HWP) rotated to 45° to map $D/A \rightarrow H/V$.
 - Then enters a PBS to separate D and A .
 - Alice: HWP2 + PBS3.
 - Bob: HWP3 + PBS5.

- After PBS, photons are coupled into multi-mode fibers (MMFs) leading to detectors.
- Detectors: Excelitas SPCMs (Single-Photon Counting Modules).
 - Detect single photons with low dark count rate ($100/s$).
- For each detection event:
 - Basis and bit value recorded.
 - Arrival time recorded.
- Data sent for post-processing to build the QKD key.

Thank you for your time!